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Examen final MN

1. Subiectul I -> Aproximare în sensul cMMP

$f(x) = 34T_0(x) + 36T_1(x) + 10T_2(x)$, $f: [-1,1] \to \mathbb{R}$

$$\begin{cases} T_0(x) = 1 \\ T_1(x) = x \\ T_2(x) = 2x^2 - 1 \end{cases} \Rightarrow f(x) = 34 + 36x + 10(2x^2-1) = 20x^2 + 36x + 24$$

Baza aproximării: $u_1 = 1$, $u_2 = x$ (polinom de grad 1)

$g = c_1 + c_2 x$

Sistem Gramm

$$\begin{cases} c_1 \langle 1,1 \rangle + c_2 \langle x,1 \rangle = \langle f,1 \rangle \\ c_1 \langle 1,x \rangle + c_2 \langle x,x \rangle = \langle f,x \rangle \end{cases}$$

$$\langle 1,1 \rangle = \int_{-1}^{1} 1 \, dx = 2$$

$$\langle x,1 \rangle = \langle 1,x \rangle = \int_{-1}^{1} x \, dx = 0$$

$$\langle x,x \rangle = \int_{-1}^{1} x^2 \, dx = \frac{2}{3}$$

$$\langle f,1 \rangle = \int_{-1}^{1} f(x) \, dx = \frac{184}{3}$$

$$\langle f,x \rangle = \int_{-1}^{1} f(x) \cdot x \, dx = 24$$

$\Rightarrow$ Sistemul devine: $\begin{cases} 2c_1 = \frac{184}{3} \\ \frac{2}{3} c_2 = 24 \end{cases} \Rightarrow \begin{cases} c_1 = \frac{92}{3} \\ c_2 = 36 \end{cases}$

$$\Rightarrow g = \frac{92}{3} + 36x$$

$$g\left(\frac{1}{2}\right) = \frac{92}{3} + 36 \cdot \frac{1}{2} = \frac{92}{3} + 18$$

$$g\left(-\frac{1}{2}\right) = \frac{92}{3} - 36 \cdot \frac{1}{2} = \frac{92}{3} - 18$$

$$\Rightarrow g\left(\frac{1}{2}\right) + g\left(-\frac{1}{2}\right) = \frac{184}{3} \in 61{,}33$$

(2.) Subiectul II → RK, ODE

$$\begin{cases} y'' = x \\ x'' = -7y \\ y(0) = 14 \\ x(0) = 21 \\ y'(0) = 0 \\ x'(0) = 0 \end{cases}$$

Notăm $a = x'$ și $b = y'$ => obținem sistemul:

$$\begin{cases} a' = x = f_1 & x' = a = f_3 \\ b' = -7y = f_2 & y' = b = f_4 \\ y(0) = 14 \\ x(0) = 21 \\ b(0) = 0 \\ a(0) = 0 \end{cases}$$

Metodă RK2 cu $\alpha = 1$

$$\begin{cases} A_0 = 0 \\ B_0 = 0 \\ x_0 = 21 \\ Y_0 = 14 \end{cases}$$

$$\begin{cases} A_{i+1} = A_i + \frac{K_{A1} + K_{A2}}{2} \\ B_{i+1} = B_i + \frac{K_{B1} + K_{B2}}{2} \\ x_{i+1} = x_i + \frac{K_{x1} + K_{x2}}{2} \\ Y_{i+1} = Y_i + \frac{K_{y1} + K_{y2}}{2} \end{cases}$$

$$\begin{cases} K_{A1} = f_1(a_i, b_i, x_i, y_i) = x_i \\ K_{B1} = f_2(a_i, b_i, x_i, y_i) = -7y_i \\ K_{x1} = f_3(a_i, b_i, x_i, y_i) = a_i \\ K_{y1} = f_4(a_i, b_i, x_i, y_i) = b_i \\ K_{A2} = f_1(a_i + K_{A1}, b_i + K_{B1}, x_i + K_{x1}, y_i + K_{y1}) = \\ = x_i + K_{x1} = x_i + a_i \\ K_{B2} = -7(y_i + K_{y1}) = \\ = -7y_i - 7b_i \end{cases}$$

$$K x_2 = a_i + K A_1 = a_i + x_i$$

$$K y_2 = b_i + K B_1 = b_i - 7 y_i$$

$$\Rightarrow \begin{cases} A_{i+1} = A_i + \dfrac{K_{A1} + K_{A2}}{2} = a_i + \dfrac{x_i + x_i + a_i}{2} = x_i + \dfrac{3}{2} a_i \\ B_{i+1} = B_i + \dfrac{K_{B1} + K_{B2}}{2} = b_i + \dfrac{-7y_i - 7y_i - 7b_i}{2} = -7y_i - \dfrac{5}{2} b_i \\ x_{i+1} = x_i + \dfrac{K_{x1} + K_{x2}}{2} = x_i + \dfrac{a_i + a_i + x_i}{2} = a_i + \dfrac{3}{2} x_i \\ y_{i+1} = y_i + \dfrac{K_{x1} + K_{y2}}{2} = y_i + \dfrac{b_i + b_i - 7y_i}{2} = b_i - \dfrac{5}{2} y_i \end{cases}$$

$$\begin{cases} A_0 = 0 \\ B_0 = 0 \\ x_0 = 21 \\ y_0 = 14 \end{cases}$$

$$\Rightarrow \begin{cases} A_1 = x_0 + \dfrac{3}{2} a_0 = 21 \\ B_1 = -7y_0 - \dfrac{5}{2} b_0 = -98 \\ x_1 = a_0 + \dfrac{3}{2} x_0 = 31{,}5 \\ y_1 = b_0 - \dfrac{5}{2} y_0 = -35 \\ x_2 = a_1 + \dfrac{3}{2} x_1 = 21 + 31{,}5 \cdot \dfrac{3}{2} = 68{,}25 \\ y_2 = b_1 - \dfrac{5}{2} y_1 = -98 + \dfrac{5}{2} \cdot 35 = -10{,}5 \end{cases}$$

$$\Rightarrow \frac{y_2 - y_1}{x_2 - x_1} = \frac{-10{,}5 + 35}{68{,}25 - 31{,}5} \approx \boxed{0{,}67}$$

3. Vectori și valori proprii + SVD

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\det(A - \lambda I_3) = \begin{vmatrix} 1-\lambda & 0 & 1 \\ 0 & -1-\lambda & 0 \\ 1 & 0 & 1-\lambda \end{vmatrix} =$$

$$= -\lambda^3 + \lambda^2 + 2\lambda = -\lambda(\lambda^2 - \lambda - 2) = -\lambda(\lambda+1)(\lambda-2)$$

$$\det(A-\lambda I_3) = 0 \iff -\lambda(\lambda+1)(\lambda-2) = 0 \iff \begin{cases} \lambda_1 = -1 \\ \lambda_2 = 0 \\ \lambda_3 = 2 \end{cases}$$

valori proprii

Pentru $\lambda_1 = -1 \Rightarrow$

$$\begin{cases} 2x + z = 0 \\ x + 2z = 0 \end{cases} \rightarrow \begin{cases} z = -2x \\ x + 2z = 0 \end{cases} \Rightarrow$$

$$\begin{cases} z = -2x \\ x - 2x = 0 \end{cases} \Rightarrow x = z = 0 \rightarrow$$

vectorul propriu asociat lui $\lambda_1 = -1$ este $v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

Pentru $\lambda_2 = 0 \Rightarrow$

$$\begin{cases} x + z = 0 \\ -y = 0 \\ x + z = 0 \end{cases} \Rightarrow \begin{cases} y = 0 \\ x = -z \end{cases} \rightarrow$$

vectorul propriu asociat lui $\lambda_2 = 0$ este $v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

Pentru $\lambda_3 = 2$ avem:

$$\begin{cases} -x + z = 0 \\ -3y = 0 \\ x - z = 0 \end{cases} \Rightarrow \begin{cases} x = z \\ y = 0 \end{cases}$$

$\Rightarrow$ Vectorul propriu asociat lui $\lambda_3 = 2$ este $v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

Raza spectrală $\rho(A) = 2$

Descompunere SVD

$$A \cdot A^T = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 2 \end{bmatrix}$$

Aflăm valorile proprii asociate matricei AA^T

$$\det(AA^T - \lambda \cdot I_3) = 0$$

$$\begin{vmatrix} 2-\lambda & 0 & 2 \\ 0 & 1-\lambda & 0 \\ 2 & 0 & 2-\lambda \end{vmatrix} = 0 \Leftrightarrow -\lambda(\lambda-1)(\lambda-4) = 0$$

$$\Leftrightarrow \begin{cases} \lambda_1 = 4 \\ \lambda_2 = 1 \\ \lambda_3 = 0 \end{cases} \Rightarrow \begin{cases} \sigma_1 = 2 \\ \sigma_2 = 1 \\ \sigma_3 = 0 \end{cases}$$

Pentru $\lambda_1 = 4$ obținem vectorul propriu $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

Pentru $\lambda_2 = 1$, obținem vectorul propriu $v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

Pentru $\lambda_3 = 0$, obținem vectorul propriu $v_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$$v_1=\begin{bmatrix}1\\0\\1\end{bmatrix} \quad \|v_1\| = \sqrt{2} \Rightarrow \quad v_1'=\begin{bmatrix}1/\sqrt{2}\\0\\1/\sqrt{2}\end{bmatrix}$$

$$v_2=\begin{bmatrix}0\\1\\0\end{bmatrix} = v_2'$$

$$v_3=\begin{bmatrix}-1\\0\\1\end{bmatrix} \quad \|v_3\| = \sqrt{2} \Rightarrow \quad v_3'=\begin{bmatrix}-1/\sqrt{2}\\0\\1/\sqrt{2}\end{bmatrix}$$

$$\Rightarrow V=[v_1' \mid v_2' \mid v_3'] = \begin{bmatrix}1/\sqrt{2} & 0 & -1/\sqrt{2}\\0 & 1 & 0\\1/\sqrt{2} & 0 & 1/\sqrt{2}\end{bmatrix}$$

$$u_1 = \frac{1}{r_1}\cdot Av_1' = \frac{1}{2\sqrt{2}}\cdot A\cdot\begin{bmatrix}1\\0\\1\end{bmatrix}=\begin{bmatrix}1/\sqrt{2}\\0\\1/\sqrt{2}\end{bmatrix}$$

$$u_2 = \frac{1}{r_2}Av_2' = \frac{1}{1}\cdot A\cdot\begin{bmatrix}0\\1\\0\end{bmatrix}=\begin{bmatrix}0\\-1\\0\end{bmatrix}$$

Pt. aflarea lui u_3 aplicăm alg. Gram-Schmidt:

$$u_3 = v_3' - \underset{u_1}{pr}v_3' - \underset{u_2}{pr}v_3' = \begin{bmatrix}-1/\sqrt{2}\\0\\1/\sqrt{2}\end{bmatrix} -$$

$$\frac{\langle u_1, v_3\rangle}{\langle u_1, u_1\rangle}\cdot u_1 - \frac{\langle u_2, v_3\rangle}{\langle u_2, u_2\rangle}\cdot u_2 =$$

$$=\begin{bmatrix}-1/\sqrt{2}\\0\\1/\sqrt{2}\end{bmatrix} - 0\cdot u_1 - 0\cdot u_2 = \begin{bmatrix}-1/\sqrt{2}\\0\\1/\sqrt{2}\end{bmatrix}$$

$$\Rightarrow \quad V=\begin{bmatrix}1/\sqrt{2} & 0 & -1/\sqrt{2}\\0 & 1 & 0\\1/\sqrt{2} & 0 & 1/\sqrt{2}\end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sigma_1 + \sigma_2 + \sigma_3 = 3$$

$$U = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & -1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

4. $$\begin{cases} S_0(x) = 1 + B(x-1) - D(x-1)^3, & 1 \le x < 2 \\ S_1(x) = 1 + b(x-2) + \frac{3\lambda}{4}(x-2)^2 + d(x-2)^3, & 2 \le x < 3 \end{cases}$$

x	1	2	3
f	1	1	20

Spline cubic natural

Cond de interpolare

$$\begin{cases} S_0(1) = 1 \\ S_0(2) = 1 \\ S_1(2) = 1 \\ S_1(3) = 20 \end{cases}$$

$$S_0'(x) = B - 3D(x-1)^2$$

$$S_1'(x) = b + \frac{3}{2}\lambda(x-2) + 3d(x-2)^2$$

Cond. de racordare

$$\begin{cases} S_0'(2) = S_1'(2) \\ S_0''(2) = S_1'(2) \end{cases}$$

$$S_0''(x) = -6D(x-1)$$

$$S_1''(x) = \frac{3}{2}\lambda + 6d(x-2)$$

Condiții la capete

$$\begin{cases} S_0''(1) = 0 \\ S_1''(3) = 0 \end{cases}$$

$\Rightarrow$ Înlocuind, sistemul devine:

$$\begin{cases} 1=1 \\ 1+B-D=1 \Rightarrow B=D \\ 1=1 \\ 1+b+\frac{3}{4}\lambda+d=20 \\ -6D=\frac{3}{2}\lambda \quad B-3D=b \\ 0=0 \\ \frac{3}{2}\lambda+6d=0 \end{cases} \Rightarrow \begin{cases} B=D \\ b+\frac{3}{4}\lambda+d=19 \\ \lambda=-4D \\ b=3D+B=4D \\ d=-\lambda\cdot\frac{3}{12}=-\lambda\cdot\frac{1}{4}=D \end{cases}$$

$$\Rightarrow b+\frac{3}{4}\lambda+d=19 \Leftrightarrow$$

$$4D-3D+D=19 \Leftrightarrow 2D=19 \Leftrightarrow D=\frac{19}{2}$$

$$\Rightarrow \begin{cases} B=\frac{19}{2} \\ \lambda=-38 \\ b=4D=38 \\ d=\frac{19}{2} \end{cases} \Rightarrow \begin{aligned} P(4)&=S_1(4)= \\ &=1+2b-28{,}5\cdot 4+\frac{19}{2}\cdot 8 \\ &=39 \end{aligned}$$

5. **Integrare numerică**

$$I_n=\int_{-1}^{1}\frac{f(x)}{\sqrt{1-x^2}}dx=\sum_{i=0}^{N}A_i\cdot f(x_i)$$

Polinoamele cebâșev are formula

$T_n(x)=\cos(n\arccos(x))$

Rezolvăm ecuația $(1-x^2)T_1'(x)=0$, unde

$T_1(x)=x \Rightarrow (1-x^2)=0 \Leftrightarrow x=\pm 1$

Rezolvăm ecuația $(1-x^2)T_2'(x)=0$, unde

$T_2(x)=2x^2-1 \Rightarrow (1-x^2)4x=0 \Leftrightarrow x=\pm 1$ sau $x=0$

$\Rightarrow I = \frac{1}{2}(I_1 + I_2) = a_0 f(-1) + a_1 \cdot \frac{1}{2} f(0) + a_2 f(1)$

Calculăm coeficienții știind că pentru $1, x, x^2$ formula este exactă:

$f(x)=1 \Rightarrow \int_{-1}^{1} 1\,dx = a_0 + a_1 \cdot \frac{1}{2} + a_2 \Rightarrow a_0 + \frac{1}{2}a_1 + a_2 = 2$

$f(x)=x \Rightarrow \int_{-1}^{1} x\,dx = -a_0 + a_2 \Rightarrow a_2 - a_0 = 0 \Rightarrow a_2 = a_0$

$f(x)=x^2 \Rightarrow \int_{-1}^{1} x^2\,dx = a_0 + a_2 \Rightarrow a_0 + a_2 = \frac{2}{3}$

$$\Rightarrow \begin{cases} a_0 = \frac{1}{3} \\ a_2 = \frac{1}{3} \\ a_1 = 2 \cdot \left(2 - \frac{2}{3}\right) = \frac{8}{3} \end{cases}$$

$\Rightarrow I = \frac{1}{3} f(-1) + \frac{4}{3} f(0) + \frac{1}{3} f(1)$

$\Rightarrow$ Integrala devine $\frac{1}{3} \cdot \frac{-93,6}{\pi} +$

$\frac{4}{3} \cdot \frac{93,6}{\pi} \cdot 0 + \frac{1}{3} \cdot \frac{93,6}{\pi} = 0$